5 November 1979

MILITARY SPECIFICATION

GAGE, SURFACE: GAGE AND PLATE

This specification is approved for use by all
Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification covers requirements of a gage (for checking the tolerance accuracy of any working surface plate) and for a master surface plate that will be used for calibration of the gage.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-P-40	Packaging and Packing of Hand Tools

MILITARY

MIL-I-18422	Indicator, Dial and Accessories

STANDARDS

FEDERAL

FED-STD-123	Marking for Shipment (Civil Agencies)

MILITARY

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of US Military Property.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in iproving this document should be addressed to AFLC CASO/LODS, Federal Center, Battle Creek, MI 49016 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5220

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(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS.

3.1 Gaging equipment. The gaging equipment required by this specification consists of the gage assembly along with the surface plate described in 3.2 through 3.2.2. The surface plate used in testing requirements of 4.4.3 is not required to be delivered along with this equipment.

3.2 Design.

3.2.1 Gage assembly design. Unless otherwise specified (see 6.2), the gage shall have the capability of indicating minimum surface plate variations of 0.000025 inches (0.00064mm) anywhere within a radial reach of not less than 9 inches (229mm) from gage assembly set-up positions. The indicator shall be provided with an adjustment feature for adjusting indicator to zero position and shall comply with requirements of MIL-I-18422.

3.2.1.1 Dial indicator. The dial indicator shall be equipped with a contact point of carbide or a material of equal or greater hardness. It shall be of shockproof construction and, unless otherwise specified (see 6.2), it shall be equipped with balanced dial readings of 0-3-0 of graduations not greater than 0.00005 inches (0.0013mm).

3.2.2 Surface plate. This gaging equipment shall be inclusive of a master granite surface plate of not less than 4 inches (100mm) thick by 18 inches (450mm) long by 12 inches (300mm) wide. Unless otherwise specified (see 6.2), this surface plate shall have an overall surface accuracy of not greater than plus or minus 0.000025 inches (0.00064mm).

3.3 Materials. Recycled and recovered raw materials should be used to the maximum extent possible in lieu of virgin raw materials as long as these materials do not jeopardize the intended use and fully comply with all contract requirements. Materials used shall be free from defects which would adversely affect the performance of maintainability of individual components or the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. None of the above shall be interpreted to mean that the use of used or rebuilt products will be allowed.

3.4 Interchangeability. All parts shall be manufactured to such standards that will permit replacement or adjustment without modification of the gage assembly.

3.5 Availability. The gage assembly requirements of this specification shall be of the latest model or configuration that is available on the commercial market for assurance of availability of replacement componets, and shall be of the design requirements and configuration as specified herein.

3.6 Construction. The gage assembly shall be complete with all necessary components and calibrating equipment so that when assembled and calibrated it will accurately perform the gaging operation for which it is designed.

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Construction shall be free from any characteristics or defects that will prevent the gage assembly from passing any of the examinations and tests described in section 4.

3.7 Equipment protection provisions. Equipment protection provisions shall be of a design of the producer and as minimum requirements shall consist of the following:

3.7.1 Surface plate. The granite surface plate supplied shall be equipped with a cover that will protect it from settling dust.

3.7.2 Gaging assembly. The gaging assembly supplied shall be provided with a protective case that would protect the gage assembly from shock.

3.8 Component condition. All parts used in fabrication of the gaging assembly shall be in as new condition and of the quality necessary to produce a gage assembly to meet the requirements described herein.

3.8.1 Castings and forgings. All castings and forgings shall be free from defects. In no event shall such processes as welding, peening, plugging, filling with cold solder or metallic pastes be used for reclaiming any part of the gage assembly and calibrating equipment.

3.8.2 Fastening devices. All screws, pins, and bolts shall be installed with means for preventing loss of tightness. All such parts, when subject to removal or adjustment, shall not be swaged, peened, staked or otherwise permanently deformed.

3.8.3 Surface finish. All surfaces shall be smooth and free from burrs and sharp edges. Surfaces that are subject to corrosion when exposed to normal use shall be protected against such corrosion in a manner that will in no way prevent compliance with this specification. The protective coating shall not loosen, chip, crack or scale with normal use and age.

3.9 Nameplate. A corrosion resistant metal nameplate, permanently and legibly marked with the following information in accordance with MIL-STD-130, shall be securely attached to each gage assembly.

Nomenclature
Manufacturer's name or identification
Manufacturer's model number
Manufacturer's serial number
Date of manufacture
Contract or order number
National stock number
US

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein,

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unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirement.

4.2 Inspection classification. Since 3.5 requires the gage assembly to be of a commercially available model, there will be no requirementes for first article inspection. Inspection requirements for each gage assembly being offered for acceptance shall consist of quality conformance inspection and tests. Failure of the gage assembly to pass comformance inspections and tests shall be cause for rejection. Quality inspections and tests shall consist of the following:

a. Examination (4.3).
b. Test (4.4 through 4.4.3).
c. Inspection, packaging, packing, and marking for shipment and storage (4.5).

4.3 Examination. Each gage assembly along with its surface plate shall be examined for design, construction, materials, components, and workmanship to determine compliance with requirements of this specification. Any deviation from these requirements shall be cause for rejection.

4.4 Tests.

4.4.1 Indicator inspection. Each indicator shall be inspected to assure proper function of all adjustments and moving parts. The spindle shall be fitted with the specified contact point (see 3.2.1.1) and shall be actuated through its entire travel range without evidence of sticking or malfunction. The dial shall be adjusted for zero calibration. For purposes of preventing unintentional movement, the provisions for adjusting the indicator for zero calibration shall either be secured in its position or shall be located in a position that would prevent its unintentional movement.

4.4.2 Master surface plate inspection. The master granite surface plate that will be part of the gaging assembly required by this specification shall be inspected and calibrated under Metrology Laboratory conditions for assurance that the surface accuracy requirements specified in 3.2.2 (see 6.2) are complied with. Failure of the gaging reference surface to conform to the specified surface accuracy shall be cause for rejection.

4.4.3 Gaging assembly functional test. The completely assembled gage shall be calibrated and subjected to the following test. On a working surface plate and by using as many different set-up positions as necessary, make at least twelve different surface measurements and check for repeatability at least three times on each of these measurements. At least four of the above measurements and repeatability checks shall be made with the extension arm in its fully extended position. Deviation of the indicator of more than one fifth dial graduation in these repeatability checks shall constitute failure of this test. The working surface plate applicable to this test is not required to be delivered along with this purchase requirement.

4.5 Inspection of preparation for delivery requirements. An inspection shall be made to determine that requirements of section 5 for preservation, packaging, packing, and marking are complied with.

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5. PACKAGING

5.1 Preservation and packaging. Preservation, packaging, and packing shall be in accordance with PPP-P-40 and as specified (see 6.2).

5.2 Marking. In addition to any special marking required by the contract (see 6.2), each unit package, intermediate and exterior container and unitized load shall be marked in accordance with MIL-STD-129. When specified (see 6.2), the marking of domestic shipments for civil agencies shall be in accordance with Fed. Std. No. 123.

6. NOTES

6.1 Intended use. Of the items required in this gaging set, the gage assembly along with its extension arm and holder is used for checking surface flatness of various surface plates. The master granite surface plate is used for checking the accuracy of and for recalibration of the gage assembly.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Gage roughness indicating capability required, if different from that specified (see 3.2.1).

c. Dial readings and graduations, if different from that specified (see 3.2.1.1).

d. Surface accuracy of surface plate supplied, if different from that specified (see 3.2.2 and 4.4.2).

e. Levels of preservation packaging, packing, and marking required (see section 5).

Custodian:
Air Force-99

Preparing Activity:
Air Force-99

Project Number:
5220-F185

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

Gage, Surface: Gage and Plate, MIL-G-87964

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

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